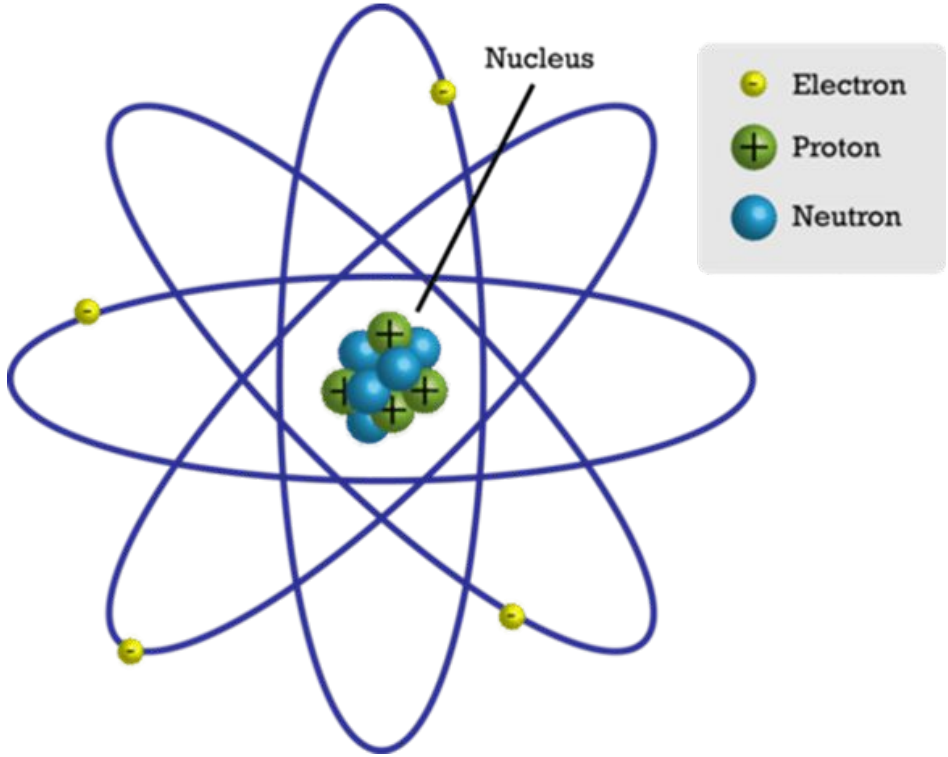
The Planetary Model

BOHR-RUTHERFORD DIAGRAMS

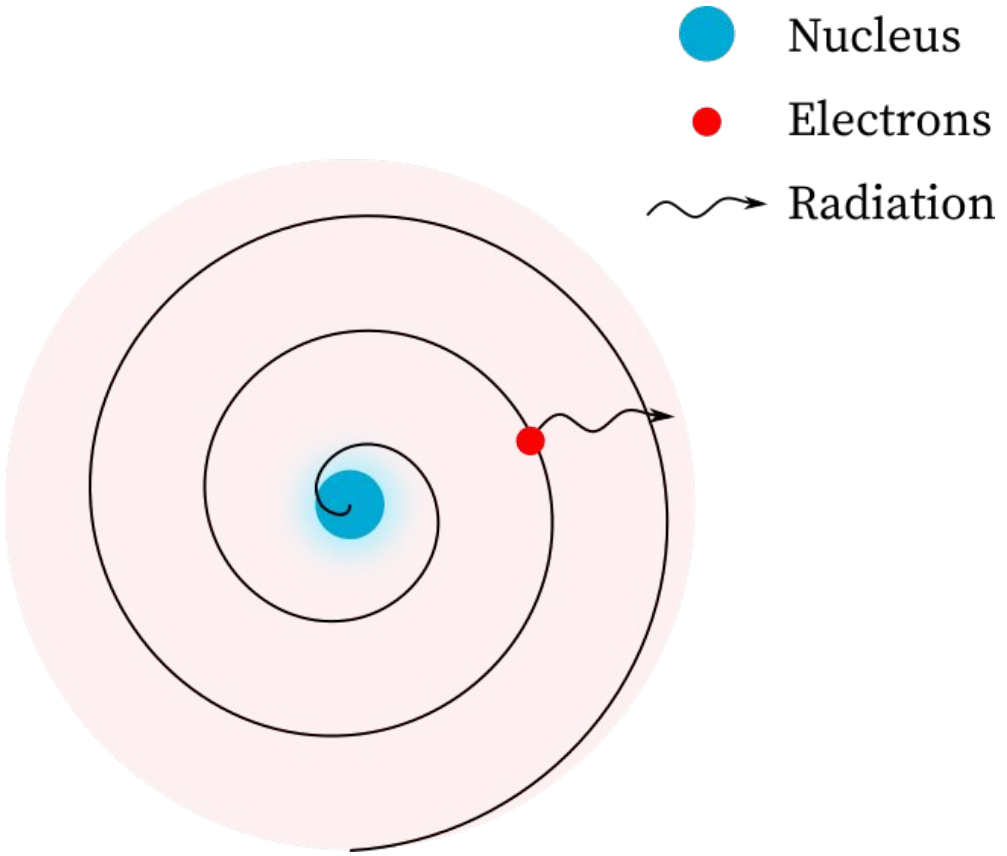
Limitations of the Nuclear Model

- The model couldn't explain where the electrons were **located**



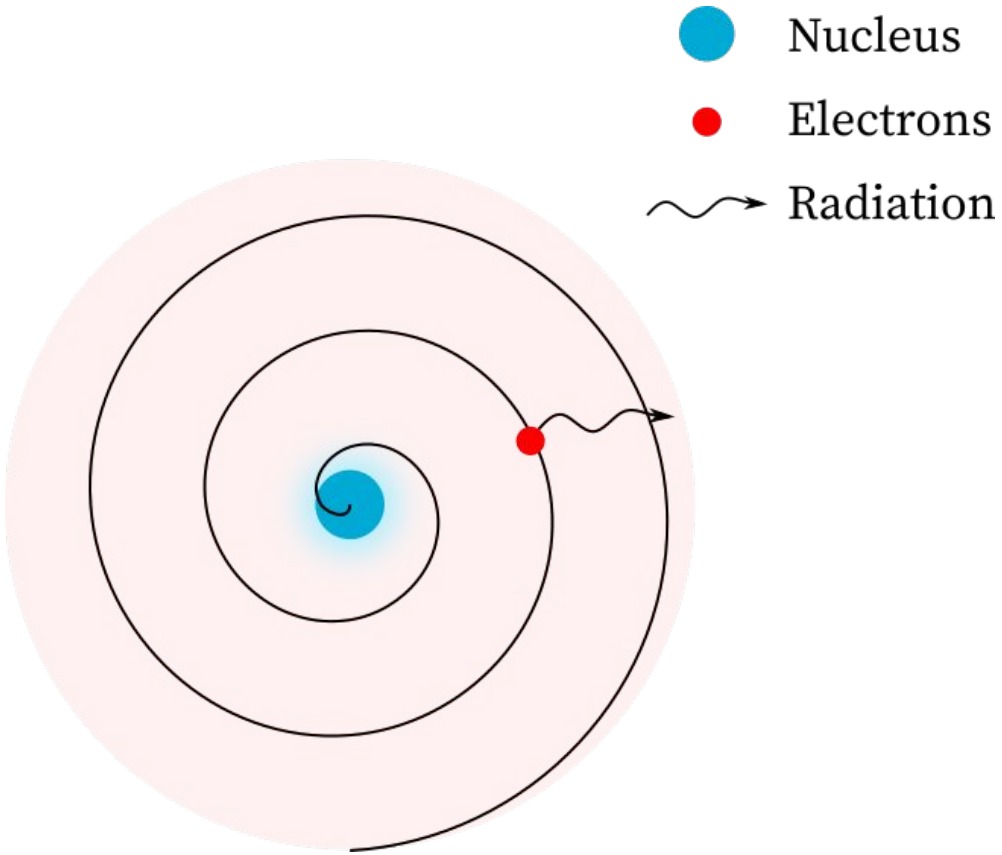
Limitations of the Nuclear Model

- Scientists at the time thought an electron in motion around a central body must continuously release **radiation** (energy)
- If an electron did this, it would lose **energy** and eventually **crash** into the nucleus destroying the atom



Limitations of the Nuclear Model

- Since this doesn't occur, the model must be **incorrect**

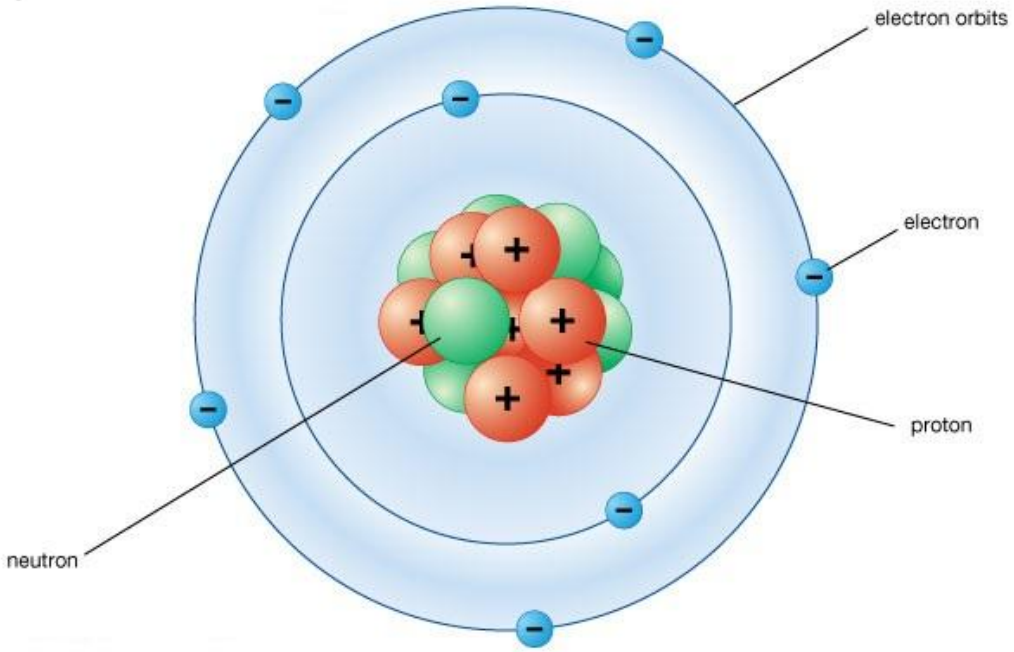


Line Spectra

Emission Spectra

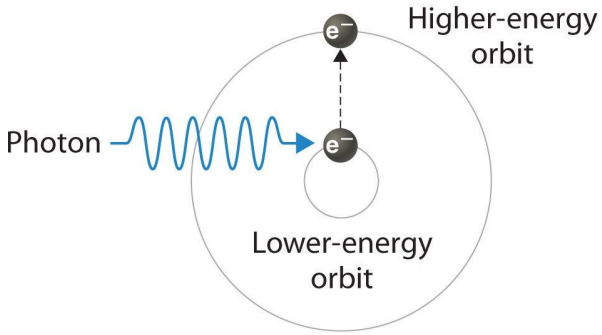
Planetary Model

- Electrons **orbit** the nucleus of an atom much like the **planets** orbit the **Sun**
- Each electron in orbit has a **definite** amount of energy
- The **further** the electron is from its nucleus, the **greater** its energy

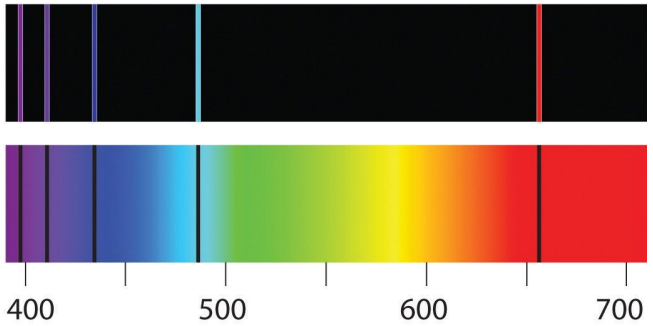


Planetary Model

- Electrons cannot be between orbits, but can **jump** to and from different orbits
- When electrons jump from higher to lower orbits, they **release** energy in the form of **light**



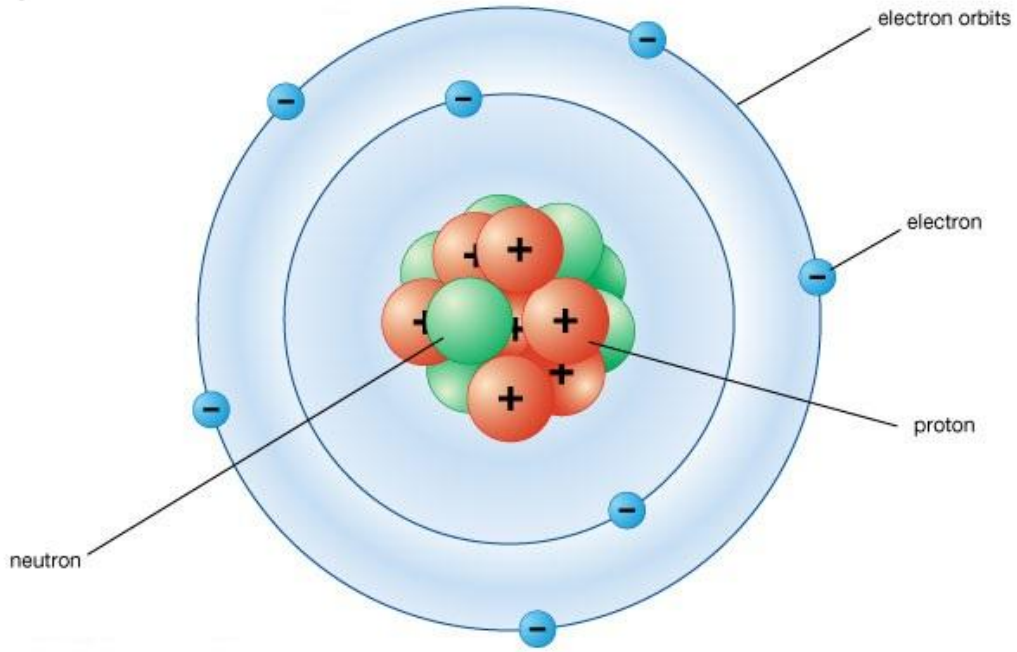
(a) Electronic absorption transition



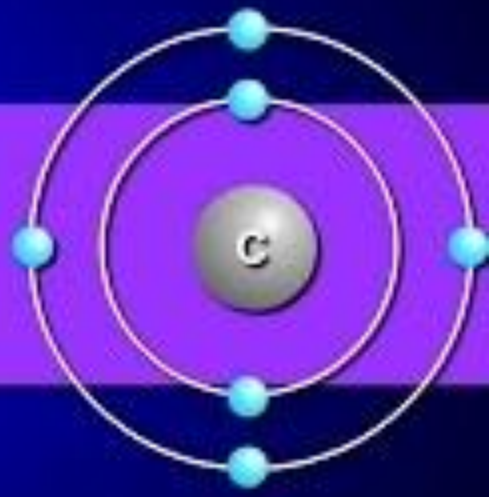
(b) H_2 emission spectrum (top), H_2 absorption spectrum (bottom)

Planetary Model

- Each orbit can hold a **maximum** number of electrons

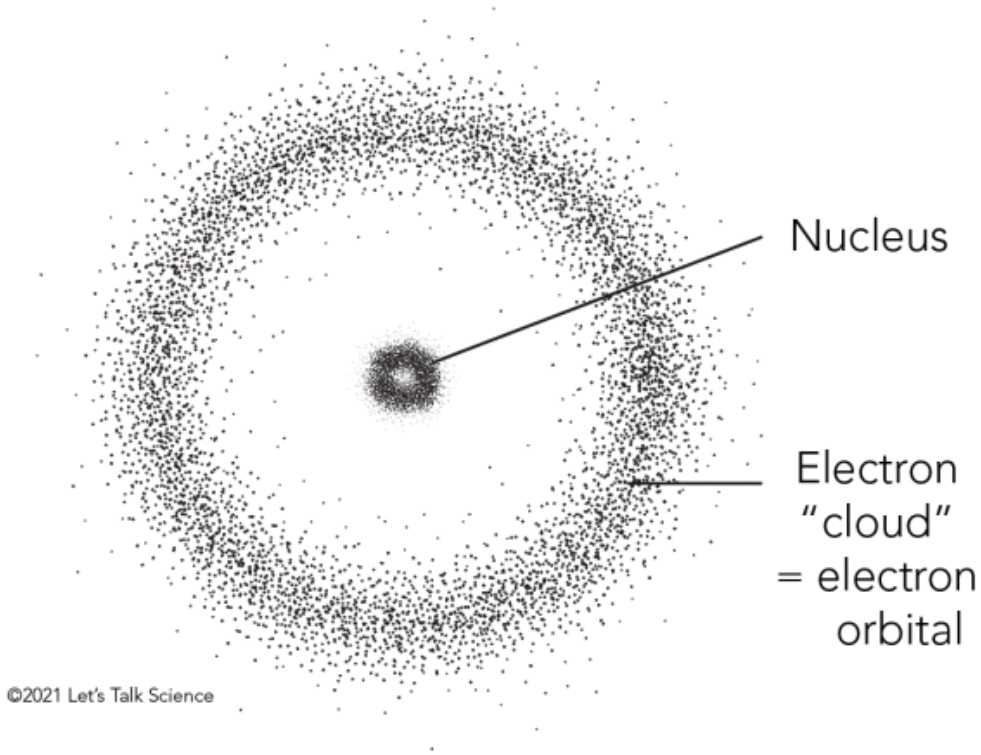


Atomic Structure



Limitations of the Planetary Model

- It could not explain the line spectra for atoms that have **two** or **more** electrons
- It would take a whole new field, called **quantum mechanics** , to explain what we currently understand about the structure of the atom

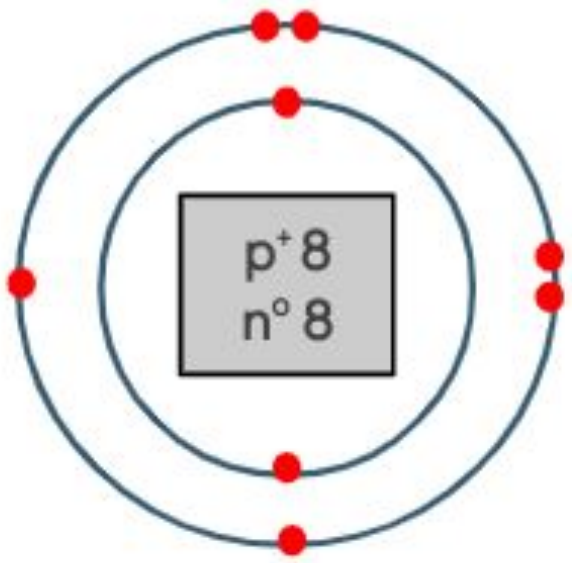


All models are wrong,
but some are useful.

George Box, British statistician (1919 – 2013)

Bohr-Rutherford Diagrams

- Each **energy level** (orbit) is shown as a ring around the nucleus
- The electrons in the **outermost** energy level are called **valence electrons**



Oxygen

Bohr-Rutherford Diagrams

- Evidence indicates that only a **limited** number of electrons can exist in each energy level

Energy Level	Maximum Number of Electrons
1	2
2	8
3	8*

Drawing Bohr-Rutherford Diagrams

1) Determine the **number** of protons, electrons, and neutrons
2) Draw the **nucleus** and indicate the number of protons and neutrons
3) Add the **electrons** one at a time, starting with the **first** energy level (the ring closest to the nucleus) and working your way out

Example #1 - Helium

Example #2 - Calcium

Your turn! (Do these steps **IN ORDER** !)

1. Bohr-Rutherford Drag & Drop
   - See Google Classroom - HAND THIS IN for Mrs. Bird to see your work!

2. Atoms Family w.s.

3. Complete the Atomic Model Summary (last row)